

AHEAD Reference Guide

Project Notes — Read Through the Lens of the DQ Automation Effort

What this document is

A re-organized summary of UNICEF's AHEAD reference guide (Mwinnyaa, Francis, Kawakatsu, Dec 2024), structured around the question driving the AHEAD automation project: *what does the current manual data quality feedback loop do, and what does an intelligent agent need to replicate, accelerate, or replace?*

The original guide is organized chronologically (demand → extraction → QA → analysis → dissemination). These notes are organized by **relevance to the automation scope**: in-scope DQ checks first, then the human judgment surfaces the agent must preserve, then the adjacent context that bounds the problem.

Scope legend

Throughout these notes, sections are tagged by their relevance to the automation work:

IN SCOPE

Mechanics the AHEAD tool will directly replace or wrap (QA flagging, feedback dialogue, decision logging).

PRESERVE

Human-judgment surfaces that must *route to people*, not be automated away (country-context interpretation, joint decisions).

ADJACENT

Upstream/downstream context (extraction, denominators, analytical products, dissemination). Bounds the problem but isn't the build.

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1. The current manual loop, in one picture

AHEAD is a UNICEF initiative producing timely subnational immunization analytics in collaboration with Ministries of Health and country EPI teams. The process spans ten stages, but the heart of it — and the part the automation effort targets — is a **two-sided feedback loop** between UNICEF HQ (which extracts data, runs QA algorithms, and produces flagged Excel files) and the country team (which reviews flags with subnational data managers and decides how to treat each one).

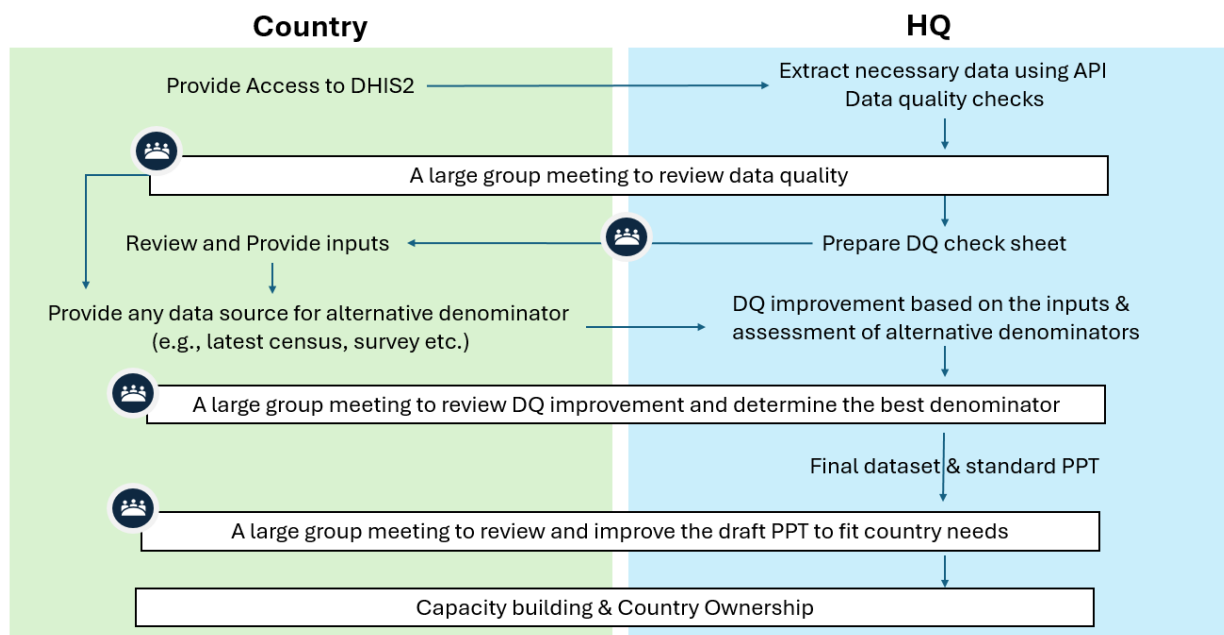


Figure (from the guide). The AHEAD process. Country-side (green) and HQ-side (blue) actions are coordinated through three large group meetings. The automation work targets the loop spanning 'Prepare DQ check sheet' through 'DQ improvement based on inputs.'

Today's loop in plain terms

- **HQ extracts** monthly facility-level data from DHIS2 (or admin2-level data from WIISE).
- **HQ runs four DQ checks** — region names, outliers, missing data, DTP1/DTP3 consistency — and produces region-specific Excel files with flagged datapoints.
- **Excel goes to country team** via email after a kickoff meeting where HQ walks through the file.
- **Country team reviews** with health-facility and subnational data managers, choosing one of a small set of responses per flag (e.g., “keep as-is, this was a campaign” vs “replace with 6-month average”).
- **Excel comes back** to HQ; UNICEF applies the cleaning decisions and produces the cleaned dataset.
- **Cleaned dataset feeds** the alternative-denominator assessment, then the analytical products (slide decks, district rankings).

What the automation effort changes

The automation tool replaces the email + Excel exchange with: (1) **flag detection at entry time**, not in batch after extraction; (2) **push notifications** via SMS, email, or voice to the right person; (3) **chat-driven correction dialogues** that capture the same decisions the Excel dropdowns capture today; (4) **cascaded flags** across facility → admin2 → admin1 → national review levels.

2. The four DQ checks — what the agent will own

Every flag the automated agent emits will trace back to one of these four checks. Each is fully specified in the reference guide with a concrete algorithm, response options, and an example Excel.

2.1 Region-name consistency

IN SCOPE

What's flagged. Admin1–admin2 region pairs whose data record count is less than the expected maximum ($n_years \times n_months \times n_vaccines$). Fewer-than-expected entries usually means the same district is recorded under inconsistent names across years.

Country-team responses. Either: (a) provide the corrected name spelling; or (b) for new or breakaway districts, name the *parent district* so historical data can be recombined.

Why automation isn't trivial. Admin boundaries change (splits, merges, renames); only the country team can confirm whether two strings refer to the same place. The agent's job is detection + a structured prompt; the human's job is naming.

2.2 Outlier detection and correction

IN SCOPE

The most algorithmically rich check. AHEAD runs **five outlier methods in parallel** and treats the count of methods that flag a point as a **confidence signal**:

- Flagged by **4+ methods** → *strong* outlier; correction decision required.
- Flagged by **1–3 methods** → *possible* outlier; review optional.

Five outlier-detection methods (operational thresholds)

#	Method	What it captures	Threshold
1	Standard Deviation (SD)	Variance from the mean.	> 2 SD from mean
2	Mean Absolute Deviation (MAD)	Average distance from mean; less affected by extremes than SD.	> 2 MAD from mean
3	Median Absolute Deviation (Median AD)	Distance from the median; robust to extremes.	> 8 Median AD from median
4	Lowess Regression	Smooth-curve fit; flags deviation from local trend.	> 2× std error from curve
5	Absolute Difference from Mean	Raw dose difference from mean.	> 100 doses higher/lower than mean

Threshold note worth flagging early

Table 1 in the guide gives **operational** thresholds (above). The **Annex** gives more conservative thresholds for the same methods (5 SD, 5 MAD, 2,500 doses). The guide doesn't reconcile this — my read is that thresholds are tuned per dataset level (annual vs monthly, admin vs facility). Either way, **thresholds should be configuration, not constants**, in the AHEAD tool.

Country-team response options (per flagged point)

- Replace with the average of doses 6 months before/after.
- No action — keep as-is (with explanation in comment).
- Treat as zero.
- Replace with a specific number.
- Use only the 'at health facility' column.
- Use only the 'during outreach' column.

admin1	admin2	unit_name	vaccine	year	month	at health facility	during outreach	Total doses	response	comment
region name	district name	health facility name	mcv1_u1	2024	3	10	38	48		
region name	district name	health facility name	mcv1_u1	2024	4	10	37	47		
region name	district name	health facility name	mcv1_u1	2024	5	10	36	46		
region name	district name	health facility name	mcv1_u1	2024	6	10	38	48		
region name	district name	health facility name	mcv1_u1	2024	7		35	35		
region name	district name	health facility name	mcv1_u1	2024	8	13	34	47		
region name	district name	health facility name	mcv1_u1	2024	9	11	350	361		
region name	district name	health facility name								
region name	district name	health facility name								
region name	district name	health facility name								

1. Replace it with the average of the doses before and after 6 months
2. No action required, Keep it as it is (please provide a short explanation in "Comment" column)
3. Treat it as 0
4. Replace to a specific number (Please add it in the comment column M)
5. Use HF data only (i.e., At_HF Column)
6. Use Outreach data only (i.e., During_Outreach Column)

The current Excel UI. Each row carries the context (admin1, admin2, facility, vaccine, year, month, facility doses, outreach doses, total). The dropdown captures the response. This is the workflow the chat interface needs to replicate.

Why country knowledge matters here. An unusually high month may be a *real* spike from a PIRI (periodic intensification activity), campaign, or outreach. An unusually low month may be a real dip from a stockout or strike. The agent can't tell these apart from the data alone — context lives with the country team.

2.3 Missing data

IN SCOPE

Computed per facility/admin2 unit. Anything below 100% completeness is flagged.

Completeness formulas

expected datapoints = # vaccines × # months × # years

Completeness (%) = (# actual datapoints / # expected) × 100

Country-team response options.

- Replace with average of data 6 months before/after.
- Replace with 0 (e.g., closed facility, no service that month).
- Replace with a specific number.
- Remove from analysis (e.g., facility was non-operational).

The first step is always recovery, not imputation. Country teams follow up with subnational data managers to find the missing report. Cleaning options only kick in if the report can't be located. An automation tool that pushes notifications at entry time should *shorten* this recovery window dramatically — the gap is smaller, the operator's memory is fresher, and the report often still exists somewhere.

admin1	admin2	facility_name	year	month	mcv1_u1	penta1_u1	penta3_u1	response	comment
region name	district name	health facility name	2022	4		11	4		
region name	district name	health facility name	2022	10					
region name	district name	health facility name	2022	11					
region name	district name	health facility name	2022	12		10	4		
region name	district name	health facility name	2023	2					
region name	district name	health facility name	2023	7					
region name	district name	health facility name	2023	8					
region name	district name	health facility name	2023	9					
region name	district name	health facility name	2023	10					
region name	district name	health facility name	2023	11					
region name	district name	health facility name	2023	12					
region name	district name	health facility name	2024	1		2	9		
region name	district name	health facility name	2024	2	9	8			
region name	district name	health facility name	2024	3					
region name	district name	health facility name	2024	4					
region name	district name	health facility name	2024	5	7	5			
region name	district name	health facility name	2024	6					
region name	district name	health facility name							

1. Replace with the average of data from six months before and after

2. Replace with 0

3. Replace with a specific number (Please directly enter specific value in each column e.g., penta1, penta3 and mcv1)

4. Keep missing data for unoperational/closed/new facilities

Missing-data Excel UI. Empty cells across vaccines (mcv1, penta1, penta3) for given months are flagged for review.

2.4 Internal consistency — DTP1 vs DTP3

IN SCOPE

The signal. DTP1 should be ≥ DTP3 in the same subnational unit (more children get their first dose than their third). When DTP3 >> DTP1, it's a likely data-entry or reporting error.

Thresholds vary by data level:

- **Annual admin2 data:** 1,000 doses and/or relative difference > 15%
- **Monthly admin2 data:** 250 doses and/or relative difference > 20%
- **Monthly facility data:** 100 doses and/or relative difference > 30%

Country-team response options.

- Keep as-is (with comment).
- Use the value of DTP1/Penta1 for both.
- Use the value of DTP3/Penta3 for both.

- Replace with specific values.
- Other (specify in comment).

admin1	admin2	facility_name	year	month	penta1_u1	penta3_u1	difference	Response	Specific value for Penta1/DTP 1	Specific value for Penta3/D TP3	Comment
region name	district b	hf 3	2021	7	44	244	200				
region name	district b	hf 7	2020	3	16	114	98				
region name	district n	hf 33	2020	3	40	135	95				
region name	district a	hf 42	2023	9	84	170	86				
region name	district p	hf 10	2023	7	9	83	74				
region name	district l	hf 40	2023	6	17	91	74				
region name	district b	hf 12	2024	8	6	77	71				

1.Keep it as it is (Please provide a short explanation in "Comment" column)
2.Use the value of DTP1/Penta1 for both
3.Use the value of DTP3/Penta3 for both
4.Replace it to specific values (Please add the values in Column L and M)
5.Others (Please specify in Comment)

Consistency Excel UI. The difference column makes the implausible gap visible at a glance.

3. Where human judgment lives — what to preserve

Reading the guide carefully, every QA step contains a sentence that amounts to: *this can only be done with country-team input*. These aren't soft suggestions — they're the parts the automation must route to humans, not bypass. Concretely:

Where	Why a human is needed
Region names	"In-depth knowledge of subnational administrative levels" needed to confirm name corrections and identify breakaway districts.
Outliers	Distinguishing <i>true</i> data-quality outliers from real spikes (PIRIs, campaigns, outreach) and real dips (stockouts, strikes) requires operational context.
Missing data	Recovery beats imputation. Only subnational data managers can chase down a missing monthly report.
DTP1/DTP3	Reconciling the gap requires understanding the local reporting process — entry error vs. legitimate revision vs. catch-up dose.
Denominator choice	The decision on which alternative denominator to use is explicitly a joint decision between HQ and country team.

Implication for design

The agent's role is **flag, route, and capture**, not **decide**. Every flagged point should reach the right human (facility data manager? district focal point? national EPI?) with enough context that they can respond in seconds. The structured response options — the Excel dropdowns — are exactly the schema the chat interface should use to log decisions back into the system.

4. Upstream context: extraction (DHIS2, WIISE)

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AHEAD requires monthly subnational/facility-level routine immunization data — vaccine doses delivered — from at least one source:

DHIS2 (or similar)

The country provides login credentials. UNICEF has an **API wrapper** that pulls monthly facility-level data. *This is the integration surface for the automation tool* — the agent will operate inside or alongside DHIS2, not as a separate system.

WIISE (WHO Immunization Information System)

Repository for routine immunization data reported through the eJRF. Pulled at admin2 level via API. Some countries restrict WIISE access; written permission may be needed for WHO to grant UNICEF access.

Once pulled, data is reformatted into a standard format before QA. **Implication for the tool:** the at-entry detection model needs the same standardized representation, but operating on individual submissions rather than batch extracts.

5. Downstream context: denominators and analytics

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Why this matters even though it's adjacent. Cleaning decisions feed directly into denominator choice and final analytics. The faster the QA loop closes, the faster downstream products refresh — this is the “why bother” payoff for automation.

Alternative denominator assessment

Coverage = doses (numerator) / surviving infants (denominator). Census-derived denominators drift over time and produce coverage > 100%. AHEAD evaluates several alternatives and validates them against DHS/MICS surveys and WUENIC.

Three derivation methods (formulas)

Redistribute national denominators

- 1) Subregion % distribution = (subregion surviving infants from national stats) ÷ (national surviving infants from national stats)
- 2) Alternative surviving infants = subregion % distribution × (national surviving infants from alternative source: WPP, IHME, WorldPop)

DTP1-derived (reverse calculation)

DTP1-derived surviving infants = (# DTP1 doses administered) ÷ (DTP1 coverage / 100)

BCG-derived (two-step, with NMR)

1) BCG-derived live births = (# BCG doses) ÷ (BCG coverage / 100)

2) BCG-derived surviving infants = BCG-derived live births × (1 – NMR/1000)

Population data sources commonly assessed

- Latest census projections (including preliminary new-census data).
- National Institute of Statistics estimates.
- **UNPD World Population Prospects (WPP).**
- **WorldPop** (University of Southampton).
- **Institute for Health Metrics and Evaluation (IHME).**
- Country-provided sources (DHS, MICS, immunization-coverage surveys).

What gets produced (the analytical packages)

Once the cleaned dataset and denominator are agreed, UNICEF produces (largely automated) slide decks and rankings:

- National decks with drill-down to admin1, admin2, and (for monthly data) facility.
- Per-admin1 decks with drill-down to admin2 and facility.
- Compilations of key slides across decks.
- **District rankings** by coverage, zero-dose, dropout, and relative change in service volume.
- Charts: coverage trends, dose changes, zero-dose distribution, unvaccinated counts, progress toward global goals — each with narrative interpretation.

6. Country ownership, cadence, advanced analytics

ADJACENT

Country ownership is the end goal

AHEAD's stated goal is **full country ownership** of the analytical tools. Products support: **planning** (priority areas), **targeted intervention** (zero-dose, dropout strategies), **fundraising** (evidence for budgets and donors), and **performance monitoring** (subnational and national progress).

Update cadence

- **Quarterly updates** for countries with frequent programmatic shifts.
- **Annual updates** for more stable environments.
- Cadence is a country-HQ joint decision; the goal is **maximum automation** of the update cycle.

Connection to the AHEAD tool: automation that shortens the QA loop also makes a quarterly cadence cheaper to run. The two efforts compound — this is part of why funders care about it.

Advanced analytics (optional, country-driven)

These aren't automation scope, but worth knowing as 'what AHEAD could integrate next' if the core DQ loop frees up bandwidth:

- **Data integration:** supply data (stockouts), outbreak data, mortality data — richer signals for prioritization.
- **Climate analysis:** seasonality, vulnerability mapping, climate-event impact on programs.
- **Physical accessibility & catchment:** travel time, road/terrain/density influence on delivery.
- **Costing simulations:** resource estimation, gap identification, cost-effectiveness of strategies.

7. Design implications for the AHEAD tool

Pulling everything above together, here are the things from the reference guide that should directly shape the tool's design:

1. The country team is the validator, not just the recipient

Every QA step in the guide depends on country-context knowledge. Outliers, missing data, DTP1/DTP3 inconsistencies all need someone who knows about the campaign in district X or the strike in region Y. **The agent must route to humans, not bypass them.**

2. Decisions are already structured — reuse the schema

Each QA check has 4–6 fixed response options — the Excel dropdowns. These are an off-the-shelf decision schema for the chat interface. Don't reinvent them; instrument them.

3. The 4+ method consensus is a built-in confidence model

Five outlier methods running in parallel, with "flagged by 4+" = *strong* and "1–3" = *possible*, is a natural triage hierarchy. Strong flags should ping someone now; possible flags can batch into a weekly review. Notification urgency should map onto this signal.

4. The cascade is explicitly hierarchical

Facility → admin2 → admin1 → national isn't just where data sits; it's the review hierarchy. A flag at facility level that isn't resolved escalates. The cascade architecture in the AHEAD vision matches this directly.

5. Thresholds are configuration, not constants

The guide gives different thresholds for annual vs monthly data, admin2 vs facility data, and even Table 1 vs Annex. Whatever the tool ships with on day one, country teams will want to tune these. Build for that from the start.

6. Recovery before imputation

For missing data, the guide is explicit: try to find the missing report first; only impute if recovery fails. The push-notification design (SMS/email/voice) directly serves this — the smaller the gap between the missing entry and the prompt, the more often recovery succeeds.

7. Closing the QA loop unblocks everything downstream

Cleaning decisions affect both the cleaned dataset and which alternative denominator wins; both feed the analytics packages. Faster QA = faster denominator agreement = fresher dashboards = more relevant decisions. This is the case to funders.